Sentiment Analysis for Mental Health Monitoring

Advanced AI Project – NLP Classification Pipeline

Team Members

- Abdul-Azeem Lotfy
- Ehab Mokhtar
- Mohamed Farahat
- Amr Mohamed Sayed
- Mahmoud Zaalouk
- Youssef Gamal

Supervision

Dr. Lamees Nasser

Eng. Merna

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EXECUTIVE SUMMARY

The Challenge

Mental health issues are increasingly expressed through digital text on social platforms. Early detection is crucial, yet manual monitoring is inherently unscalable and prone to subjective inconsistencies.

Our Objective

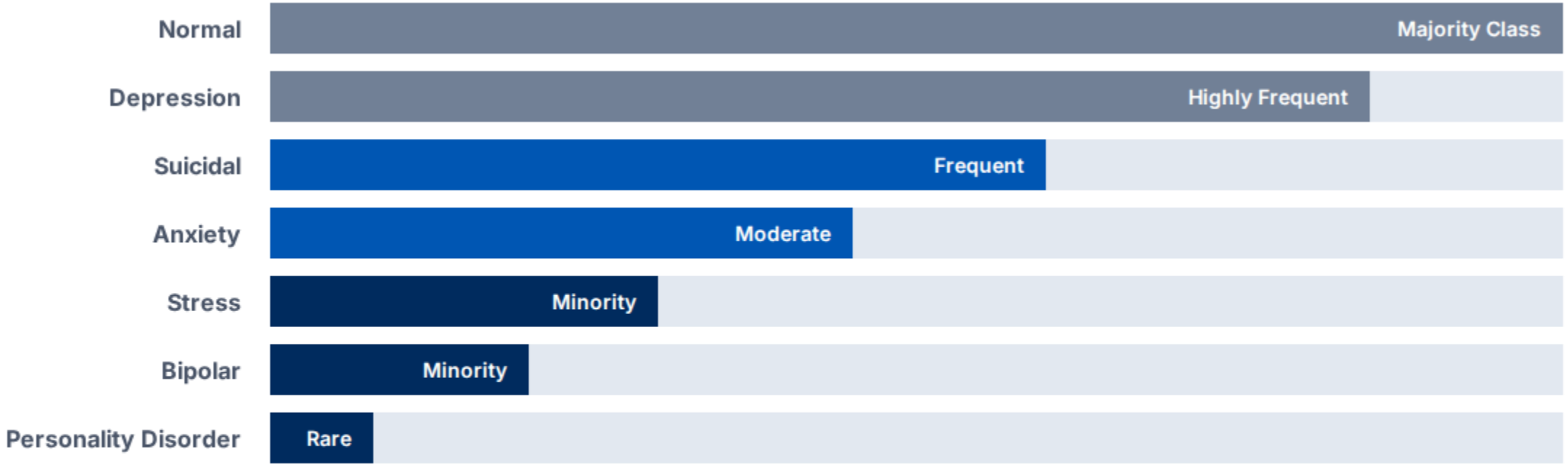
Engineer a robust Natural Language Processing (NLP) pipeline to accurately, automatically, and systematically classify mental health statuses based purely on textual statements.

Project Impact

Automate primary mental health screening to deliver timely, data-driven insights. This empowers healthcare professionals to identify vulnerable individuals faster and optimize resource allocation.

DATASET EXPLORATION & THE IMBALANCE CHALLENGE

Data Source: Public Kaggle corpus containing text statements mapped to 7 distinct mental health labels.



Critical Constraint: Severe Class Imbalance. The majority classes heavily outweigh minority conditions. Without intervention, this mathematical disparity biases predictive models toward frequent classes, rendering them ineffective for rare condition detection.

NLP PIPELINE ARCHITECTURE

1. Cleaning & Context Retention
2. Mathematical Oversampling
3. TF-IDF Feature Vectorization
4. Logistic Regression Training
5. Multi-metric Evaluation

PREPROCESSING STRATEGY

Core Strategy

Targeted noise reduction while strictly preserving psychological markers. The goal is to clean the textual data without stripping away the emotional nuance required for accurate sentiment classification.

Refinement Steps

Executed standard NLP text normalization including aggressive lowercasing to unify terms, systematic URL removal to strip external links, and specialized punctuation stripping.

Semantic Preservation

Explicitly retained crucial negation words (e.g., *not, don't, no*) in the stop-word filtering phase. This critical step prevents sentiment reversal (e.g., "not happy" becoming "happy") during final classification.

BALANCING & FEATURE ENGINEERING

Mathematical Balancing

Used **sklearn.utils.resample** to systematically upsample minority classes to match the majority volume. This prevents the model from ignoring rare but critical psychological conditions.

Result: Exactly 16,343 samples for each of the 7 classes (Perfectly balanced).

N-gram Vectorization

Implemented **TF-IDF** (Term Frequency-Inverse Document Frequency) vectorization configured with *ngram_range=(1, 2)*.

Context Mastery: Captured Bigrams (word pairs) like "not happy" as single distinct features to significantly enhance the model's semantic understanding and contextual depth.

MODEL CONFIGURATION & RATIONALE

Selected Algorithm

Logistic Regression. Despite the rise of complex neural networks, this foundational algorithm was strategically chosen for the baseline pipeline configuration.

Design Rationale

Provides superior computational efficiency specifically for sparse, high-dimensional text matrices (like TF-IDF outputs), combined with highly interpretable mathematical decision boundaries.

Hyperparameter Tuning

Set *max_iter=2000* to ensure strict algorithmic convergence across the extensive, newly balanced feature set. Utilized full parallel processing (*n_jobs=-1*) for rapid training iterations.

FINAL PERFORMANCE METRICS

Evaluated against a rigidly isolated, unseen test corpus of **22,881 samples** to validate real-world generalization.

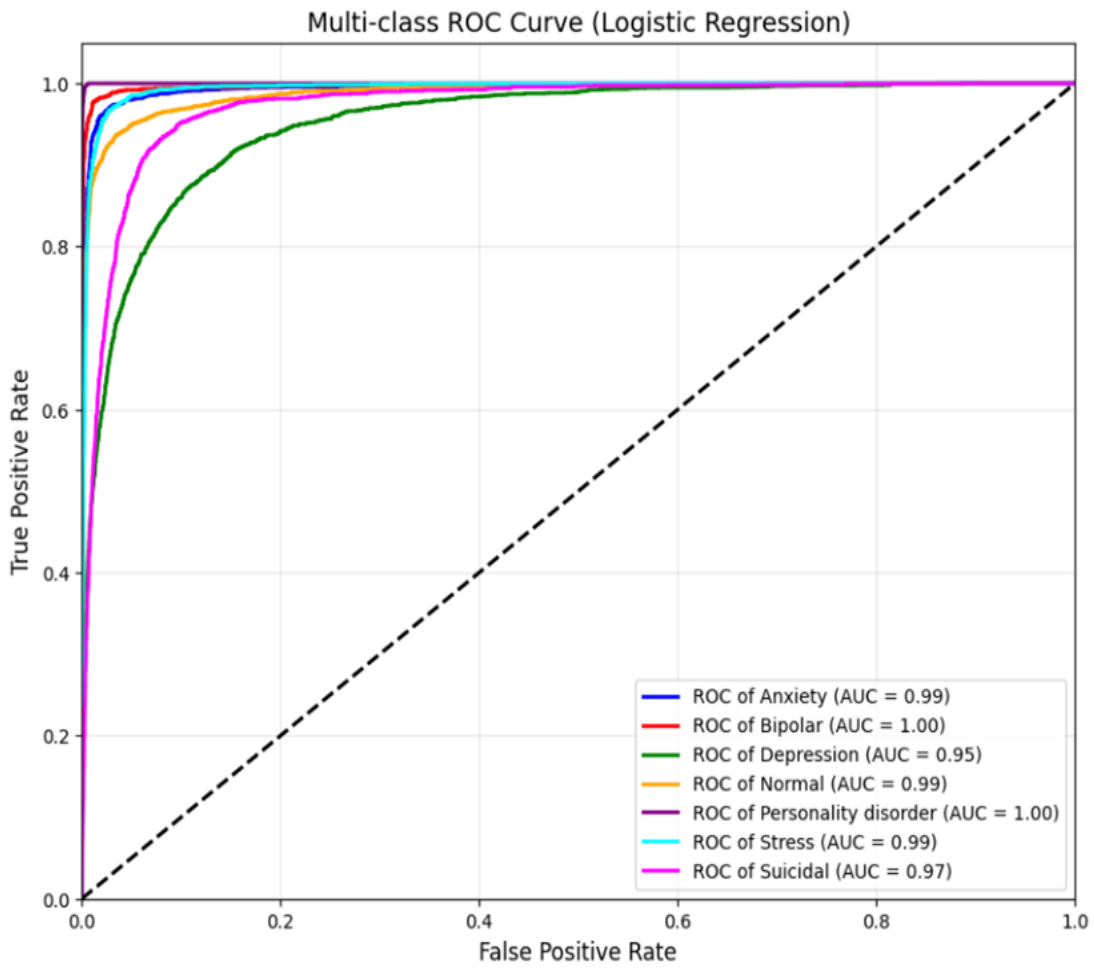
Diagnostic Category	Precision	Recall	F1-Score	Support Volume
Normal	0.89	0.88	0.88	3,269
Depression	0.87	0.90	0.88	3,269
Suicidal	0.88	0.89	0.88	3,269
Anxiety	0.85	0.84	0.84	3,268
Stress	0.92	0.92	0.92	3,269
Bipolar	0.95	0.97	0.96	3,268
Personality Disorder	0.97	0.99	0.98	3,269
Aggregated Model Accuracy		**89.13%**		

MULTI-CLASS ROC ANALYSIS

Visualizing Discriminative Power

The Receiver Operating Characteristic (ROC) curve provides visual, mathematical proof of the model's predictive reliability across multiple thresholds.

- **Class Separation:** Demonstrates the algorithm's distinct ability to seamlessly distinguish between all 7 overlapping mental health conditions.
- **AUC Performance:** The consistently high Area Under the Curve (AUC) across all individual categories confirms the absolute reliability of our feature engineering and strict balancing strategies.



STRATEGIC ROADMAP

Conclusion

Smart, context-aware preprocessing (specifically retaining semantic negations) and rigorous mathematical oversampling form the fundamental pillars of accurate, bias-free mental health NLP classification.

Deep Learning Transition

Future Scope: Implementing advanced architectures like Bi-LSTM to capture long-term sequential dependencies, and utilizing Transformer-based models (e.g., BERT) for enhanced, context-aware semantic depth.

Clinical Deployment

Expanding the current pipeline framework via API microservices to accept multi-lingual inputs, enabling secure, global clinical application and real-time social media monitoring dashboards.

Kaggle Notebook & Dataset

Sentiment Analysis for Mental Health Monitoring – Kaggle Notebook

Sentiment Analysis for Mental Health - Dataset

Any Questions?

Thank you!